Jordan L Vorster

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GitHub: github.com/golba98 | Portfolio: jordanvorsterwebsite.online

PROFILE SUMMARY

I'm a Computer Science student at the University of London, and most of my time goes into LLMs and the tools around them. I built Ubume, a terminal app where coding agents and local models plan, call tools and wait for approval before they change anything. I've also built LLMs from scratch: one in NumPy with my own autograd engine, and a 900M-parameter one in PyTorch with its own tokenizer and conversation memory. I want to keep learning this alongside people who build these systems for work.

EDUCATION

University of London | *BSc Computer Science, full-time distance learning* *Currently enrolled*

PROJECTS

Ubume CLI | *TypeScript, Bun, Ink* github.com/golba98/Ubume-CLI

- Published an npm package that runs Codex, Claude Code, Antigravity, Mistral Vibe and local models in one terminal UI, with sessions that save, resume and rewind
- Built the human-in-the-loop controls: plan review before anything runs, four permission modes from read-only to full access, and a diff of every file change
- Connected local models (LM Studio, Ollama, Unsloth) to an agent runtime with tool calling, Playwright browser tools and approval before each click or form entry

LLM From Scratch (PyTorch) | *Python, PyTorch* github.com/golba98/LLM-From-Scratch

- Built and trained a 900M-parameter decoder-only Transformer in PyTorch across eight packages: BPE tokenizer, data pipeline, training with checkpoint recovery, and inference
- Implemented conversation memory with SQLite storage and cosine-similarity retrieval over embeddings, plus classifier heads on a frozen embedding encoder

Codexa NumPy | *Python, NumPy, CuPy/CUDA* github.com/golba98/Codexa-NumPy

- Built and trained a language model in NumPy with my own tensor/autograd engine, attention, RMSNorm, SwiGLU, cross-entropy and AdamW, plus a CuPy backend for NVIDIA GPUs
- Scaled to 253M parameters with CPU/GPU parity tests, training at 4,000 tokens/sec on an RTX 4080

SyncroEdit | *JavaScript, Cloudflare Workers, Yjs* syncroedit.online

- Built a real-time collaborative editor with Yjs CRDTs over WebSockets, a Durable Object per room, D1 (SQL) persistence and Playwright convergence tests

HP Prime Emulator | *Rust* github.com/golba98/hp-prime-emulator

- Built a compiler and VM for HP Prime programs: Pratt parser, bytecode compiler, stack VM, desktop IDE with a step debugger, and USB testing against the real calculator

HTTP Server | *C++23* github.com/golba98/HTTP-Server

- Wrote an HTTP/1.1 server on POSIX sockets: incremental RFC 9112 parser, sandboxed static files and a bounded thread pool, tested with sanitizers and libFuzzer

IT Pixel Forger | *C++23, Qt 6* github.com/golba98/IT_Pixel_Forger

- Rewrote my Python image-mosaic tool in C++23, making it about 12x faster, with a Qt 6 GUI and animated GIF export

More on GitHub: 3D Portfolio (TypeScript, React, Three.js), MyGUI (C++20 GUI library on OpenGL) and Forest RPG (procedural world generation in JavaScript)

WORK EXPERIENCE

Part-time Waiter *2022*

- Served customers in a busy, high-pressure restaurant

Summer Crew Member *2021, 2023*

- Prepared rooms and maintained guest and pool areas during peak holiday seasons
- Led a small team to have all guest areas ready on time and to standard

Part-time Intern | *Pam Golding Properties* *2024–2026*

- Photographed homes and prepared listings and documentation for new properties
- Assisted with inventory checks, property value estimates and preparing houses for arrivals
- Worked with agents to troubleshoot issues and streamline daily operations

LEADERSHIP & INVOLVEMENT

Debating Society *Grade 8–12*

Chess Team | *Represented the school in inter-school competitions*

Programming Team | *Co-Leader, mentored peers on team projects*

South African Mathematics Olympiad (SAMO) | *Participant* *Grade 9–11*

CERTIFICATIONS

- **CS50: Introduction to Computer Science** | *Harvard University (edX)*: C, Python and algorithms
- **Introduction to Cloud Computing** | *Coursera*
- **Introduction to Calculus / Calculus Fundamentals** | *Coursera*

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, C++, Rust, C#, Delphi, SQL
- **AI & Data:** PyTorch, NumPy, CuPy/CUDA, Transformer architecture, BPE tokenization, embeddings, cosine-similarity retrieval, SQLite
- **Agents & LLM Apps:** Tool calling, human-in-the-loop approval, permission modes, session state, conversation memory, embedding retrieval, OpenAI-compatible and Anthropic APIs, JSON
- **DevOps & Tools:** Git, GitHub Actions (CI), Docker, Podman, Cloudflare (Workers, D1, Durable Objects, Wrangler), Render, Vercel, Linux (Fedora), CMake, uv, conda, Playwright, Vitest, pytest
- **Web:** HTML, CSS, React, Vue, Angular, Node.js, Express, Bun, Electron, Three.js
- **LLMs & AI Tools:** vLLM, llama.cpp, MLX, Ollama, LM Studio, Unsloth, Hugging Face TRL

INTERESTS

- Building things, from software tools and games to custom PCs
- Running, benchmarking and fine-tuning local LLMs
- Maths

SOMETHING I BUILT USING AI

I use AI coding agents (Codex, Claude Code and Gemini) on my HP Prime emulator, a Rust compiler and VM for calculator programs. I wrote an AGENTS.md that explains the project, its crates and its commands, so the agents know how to build and test it. I don't just trust what they write. Changes have to pass the Rust tests in CI, and I check behaviour with differential tests that run the same program on the emulator and on my real HP Prime and compare the output.

WHAT INTERESTS ME ABOUT INTELLIGENT AGENTS

What interests me is the science behind it. Under the hood, an LLM is just doing probability calculations, billions of times a second. But at the end of that, it can write an entire small JavaScript project the way a person would. I think that says a lot about us too. We started from something as simple as DNA, and got to the point where we can question and reason about reality. When I build agents, I get to see how far that goes.